Practical Estimation of R_0 from various sources of information

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Contents

Before going further: Download the zip-file "EstimationR0FieldData.zip" from the website and unzip at a convenient location.

1. Estimation of R_0 from the exponential growth phase

Theory

When an epidemic is in its initial phase, the number of infected individuals (prevalence) increases (more or less) exponentially, as does the number of new cases per unit of time (incidence). Because during the exponential growth phase the proportion of susceptibles in the population remains approximately constant ($X \approx N \Leftrightarrow x \approx 1$, if everyone is susceptible), it is possible to estimate the exponential growth rate r from incidence data, and then use the exponential growth rate r to estimate R_0.

What we need to link r and R_0 is the mean *generation time* T_g, which is the mean time at which an infected person infects new cases, measured from the moment she was infected herself. In the basic SIR model, the mean generation time is equal to the mean duration of the infectious period, so $T_g = 1/\gamma$. From what we have seen in previous day, we can derive the relation between r and R_0 as follows:

$$\left.\begin{array}{l} R_0 = \dfrac{\beta}{\gamma} \\ r = \beta - \gamma \end{array}\right\} \Rightarrow R_0 = 1 + \frac{r}{\gamma} = 1 + rT_g$$

The dataset and analysis

Open the file "SARS.csv". You see a dataset, derived from the Hong Kong SARS epidemic, but changed by some random alterations. The dataset contains the daily incidence for 655 cases that were infected until the 55th day of the epidemic, and that were not linked to some particular clusters, so that mixing can be assumed to have been more or less random.

(a) how many cases were infected in the first, second, and third week of the epidemic?

(b) what is the observed incidence on day 2?

Close the file "SARS.csv" without saving. Then open file "EstimationR0FieldData.R" in R.
Study the lines 1 to 34 and try to understand what is done in each line.

At line 30 set the working-directory (setwd) to the directory containing the material that you have extracted from the "EstimationR0FieldData.zip" file.

Run lines 1 to 34 (you might need to install the libraries "graphics" and "bbmle").

To estimate the exponential growth rate r, we only need the data until the growth of the epidemic is not exponential anymore.

(c) by visual inspection of the plotted graph, when would you think the exponential growth phase ends?

Log-transforming the y-values makes it easier to see when the exponential growth phase ends. An exponential curve e^{rt} becomes linear when plotted on a logarithmic scale, because $\log(r^{rt}) = rt$.

- replace `y = sars$Incidence` by `y = log(sars$Incidence)`

(d) by visual inspection, when does the exponential growth phase end?

On line 37 set `end.exp.phase <- 20` to the last day of the exponential growth phase you decided on.
According to the model we expect an exponential curve described by the equation $i(t) = i(0)e^{rt}$. We will now use maximum likelihood to estimate $i(0)$ and r. We will use a Poisson model, which means that we assume that the observed incidences $c(t)$ were sampled from a Poisson distribution, with means equal to the expected incidences $i(t)$. Thus, the probability to observe incidence $c(t)$, is

$$\Pr\left(C(t)=c(t)\middle|r=\ldots,i(0)=\ldots\right)=\frac{e^{-i(t)}i(t)^{c(t)}}{c(t)!}$$

Where this equation calculates the *probability* of observing 0, 1, 2, ... cases *given* values for r and $i(0)$, it can also be used to calculate the *likelihood* of parameter values for r and $i(0)$ *given* an observed number of cases:

$$likelihood\left(r,i(0)\middle|C(t)=c(t)\right)=\frac{e^{-i(t)}i(t)^{c(t)}}{c(t)!}$$

This log-likelihood is calculated in lines 45 to 48 in which we define a function `dll`

The likelihood of r and $i(0)$ for all observations together is the product of all these single likelihoods. This means that the *log*-likelihood for all observations is the *sum* of all single *log*-likelihoods.

We next want to change r and $i(0)$ such that the log-likelihood is as high as possible. The r and $i(0)$ that result in the highest log-likelihood are the *maximum likelihood estimates*. Therefore we use the function `mle2`.The result is stored in variable `fit`.

(e) What is the value of the log-likelihood? What is the estimated r?

Let us check if we decided right on which part of curve was the exponential phase.

Run lines 58 – 68 to plot a bright red line for the exponential phase and a pink line to extrapolate the fitted curve.

(f) Was the selected end of the exponential phase chosen correctly?

Our final step is to estimate R_0. We can use the equation given in the introduction. The mean generation time is in the range 10-16 days.

(g) What is the range of R_0 values for the possible generation times?

2. Estimation of R_0 from the final size of an epidemic

Theory

The final size of an epidemic represents the fraction of the population that has experienced infection by the end of an epidemic. Comparing final sizes of 30% and 90%, we feel that a disease with 90% has the higher transmissibility than that with 30%. Indeed, the final size is directly relevant to the transmission potential of an infectious disease, and is determined by the next-generation matrix and also by R_0. In a homogeneously mixing population, we have the so-called final size equation:

$$1 - z(\infty) = \exp(-z(\infty) R_0)$$

where $z(\infty)$ is the final size ($0 < z(\infty) < 1$), the proportion of the population infected during the outbreak. From this equation, the following estimator for R_0 can be derived:

$$R_0 = \frac{-\log\left(1 - z(\infty)\right)}{z(\infty)}$$

This is a general estimator for homogeneously mixing populations. For instance, it is independent of the distribution of the generation time, unlike the estimation of R_0 from exponential growth.

The dataset and analysis

Open the file "RacingHorses.csv". You see a dataset, derived from an epidemic of influenza A virus (H3N8) among racehorses in Japan. The dataset contains the population size of five race courses and the number of horses that got infected during the outbreak.

(a) what was the final size of the outbreak in Niigata?

Close "RacingHorses.csv" without saving. Run lines 87-89 of "EstimationR0FieldData.R" in R. You see a barplot of all final sizes. The next step is to calculate the confidence intervals of the final sizes. We do this by the normal approximation to the binomial distribution. This means that we will have to calculate a standard error, so that we can calculate $p \pm 1.96SE$ as the 95% confidence interval.

(b) how do you calculate the standard error for a proportion?

Line 93 contains the formula to calculate the standard error. Does it match your answer?

Run lines 91-96. The figure now shows the final sizes and the 95% confidence intervals for the five race courses. The next step is to use the estimates and confidence intervals to calculate estimates and intervals for R_0.

Run lines 98 – 108 to calculate R_0 and its confidence intervals. Do you understand the code?

(c) Which race course had the highest R_0? Which race courses have significantly different R_0 values, based on the confidence intervals?

3. Estimation of R_0 from the endemic steady state

Theory

When an infection is endemic in a population, the proportions of susceptible, infected, and recovered animals stay more or less the same. In the basic SIR model, with birth and mortality,

$$\frac{dx}{dt} = \mu - \beta xy - \mu x$$
$$\frac{dy}{dt} = \beta xy - \gamma y - \mu y$$
$$\frac{dz}{dt} = \gamma y - \mu z$$

this results in the following steady state proportion of immunes:

$$z^* = \frac{\gamma}{\mu + \gamma}\left(1 - \frac{1}{R_0}\right)$$

Now, if we would observe z^* (the seroprevalence, if all immunes are seropositive), we can estimate R_0 as:

$$R_0 = \frac{1}{1 - z^* \frac{\mu+\gamma}{\gamma}}$$

For many infectious diseases with life-long immunity, the recovery rate γ is much bigger than the death rate μ ($\gamma >> \mu$). That means that $(\mu + \gamma)/\gamma$ is almost equal to 1, so that R_0 can be estimated as $1/(1 - z^*)$.

For our example below, Salmonella Dublin in dairy cows, we will look at a bit more complicated situation, because cows lose immunity not very long after infection. We also include a latent period into the model, but we neglect mortality because infection and loss of immunity occur at much faster rates than birth and death. This results in the model:

$$\frac{dx}{dt} = \eta z - \beta xy$$
$$\frac{dw}{dt} = \beta xy - \sigma w$$
$$\frac{dy}{dt} = \sigma w - \gamma y$$
$$\frac{dz}{dt} = \gamma y - \eta z$$

The steady state proportion of immune individuals can be derived as

$$z^* = \left(1 - \frac{1}{R_0}\right) \frac{\gamma\sigma}{\gamma\sigma + \eta\sigma + \eta\gamma} = \left(1 - \frac{1}{R_0}\right) \frac{D_S}{D_S + L + D}$$

In this equation, $R_0 = \beta/\gamma$, D_S is the mean duration of seropositivity, L is the mean latent period, and D is the mean infectious period.

(a) how would you express D_S, L, and D in terms of the model parameters β, σ, γ, and η?

From the steady state seroprevalence, we can derive the following estimator for R_0:

$$R_0 = \frac{1}{1 - z^*\left(1 + \frac{L+D}{D_S}\right)}$$

The dataset and analysis

Open file "Salmonella.csv". You see a dataset from two dairy herds in the Netherlands, that were sampled on several occasions for the seroprevalence of Salmonella Dublin (Van Schaik et al 2007; Vet Res 58, pp 861-9). For each sampling day, the number of cows is listed as well as the number of seropositive cows. Time is measured since the first sample.

(b) what are the final seroprevalences in both herds, and what were the time intervals between the first sample and these observations?

Close file "Salmonella.csv" without saving. Run lines 121-124 of "EstimationR0FieldData.R" in R. The plot shows the course of the seroprevalence in both herds.

(c) by visual inspection, when does the endemic steady state start for the two herds?

We will now estimate the endemic steady state and R_0 for the two herds.

On lines 127-128 you can indicate the moment on which the endemic steady states starts. The mean seroprevalence in the steady state can be calculated using lines 131-132. Do you understand the code?

(d) what were the seroprevalences in the endemic steady states for the two herds?

Plot the steady states using lines 135-136.

(e) do you still agree on the starting moment of the steady state

From literature it was found that the latent period is about 2 days, the infectious period about 10 days, and that cows are seropositive for about three months. Run lines 138- 147 and a resulting estimate for R_0 appears.

(f) what was R_0 in the two herds?

Bonus

(g) How sensitive is the estimate of R_0 to the values found in literature of the latent period, the infectious and the time during which cows are seropositive?